AcadMate Application and Installation Manual

F2026 Team 3 | CS 691 Sprint 2 | 7 November 2026

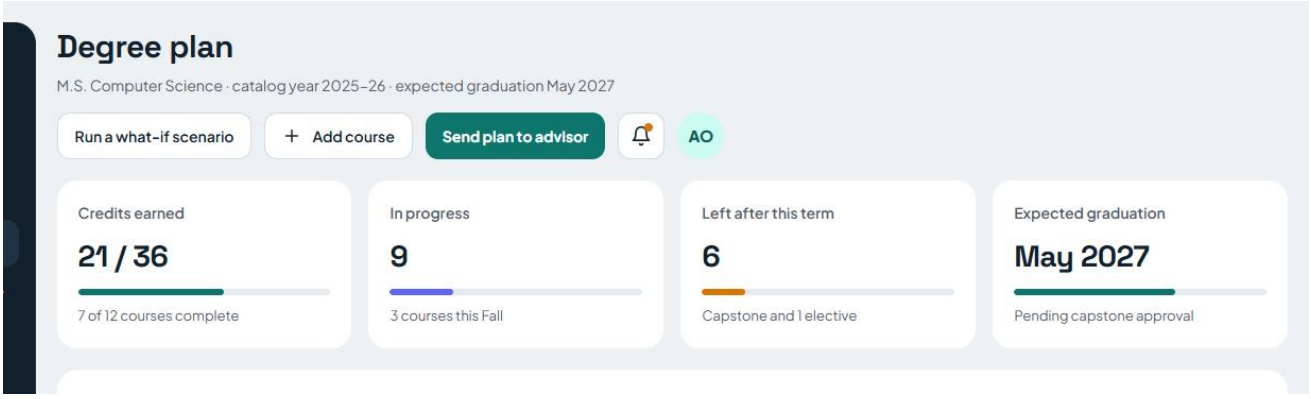
1 Sign in and inspect degree progress

Open the application and sign in with the account for your role. For the existing isolated demonstration, use http://localhost:55173. A fresh local installation uses http://localhost:5173. Keep student and advisor sessions in separate browser profiles or browser contexts.

Role	Synthetic demo email	Home page
Student	amara.okafor@pac.edu	/student
Advisor	sarah.smith@pac.edu	/advisor
Administrator	jordan.lee@pac.edu	/admin

The seeded demonstration password is Acadmate!2026. These accounts contain invented academic records. Use newly provisioned credentials for any real installation.

As Amara, open Degree plan from the sidebar. The checked record shows GPA 3.73, 21 of 36 credits earned and nine credits in progress. Six credits remain after the current term. Foundation requirements are 2 of 2 and core requirements are 4 of 4. Electives show one of five completed with three in progress. The capstone remains open.

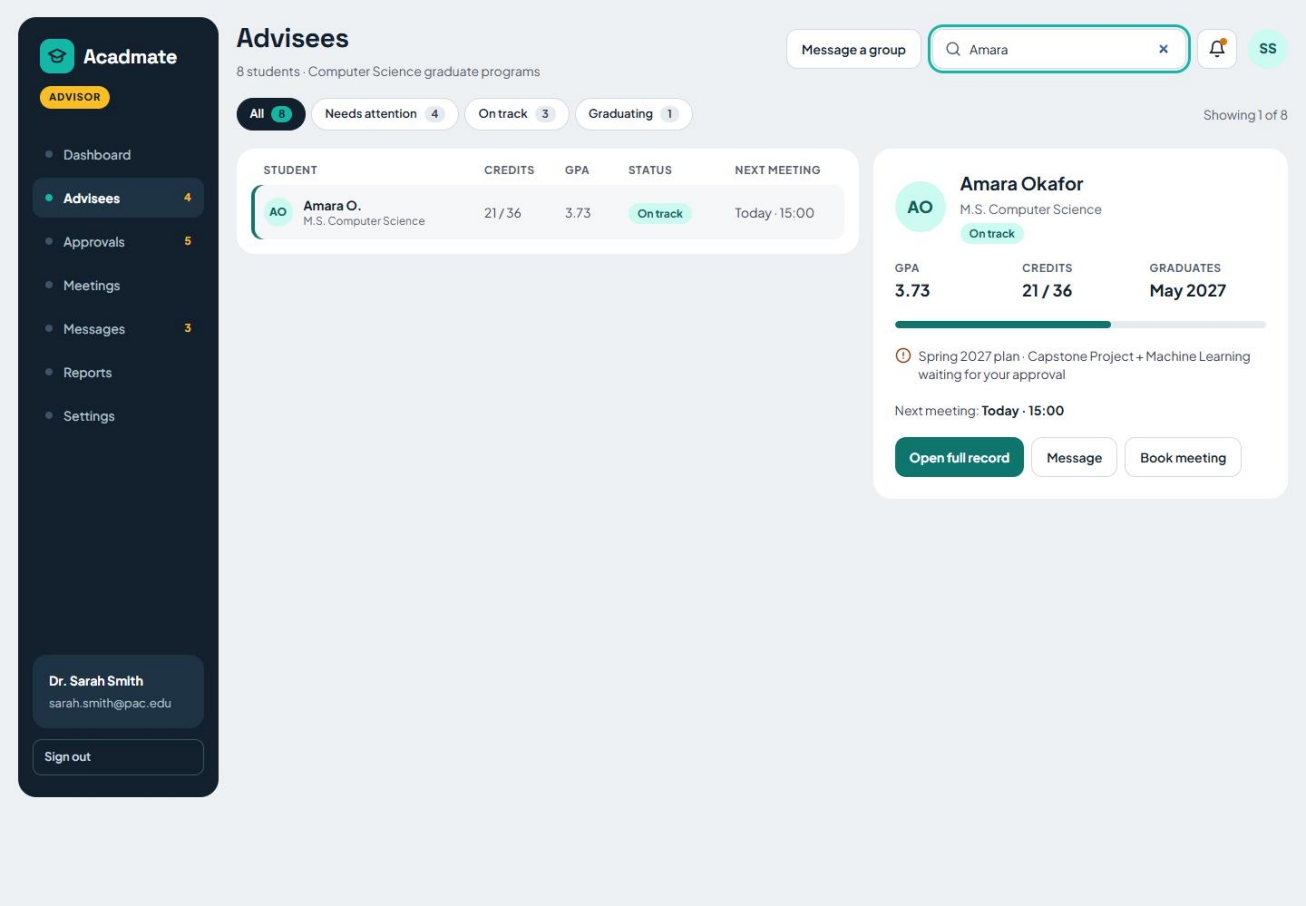


Degree progress summary for the seeded student.

2 Submit a plan and review it as an advisor

In Degree plan, review the proposed Spring 2027 courses. The demonstrated request contains CS 680 Machine Learning and CS 691 Capstone Project. Resolve any prerequisite or offering error before submitting. After submission, the course rows show Pending approval.

Sign in as Dr. Smith in the other session. Open Advisees and enter Amara in the search field. One student should remain from the eight advisees. Clear the field to restore the list. A status filter and a name query apply together, so a valid query can still produce an empty filtered result.

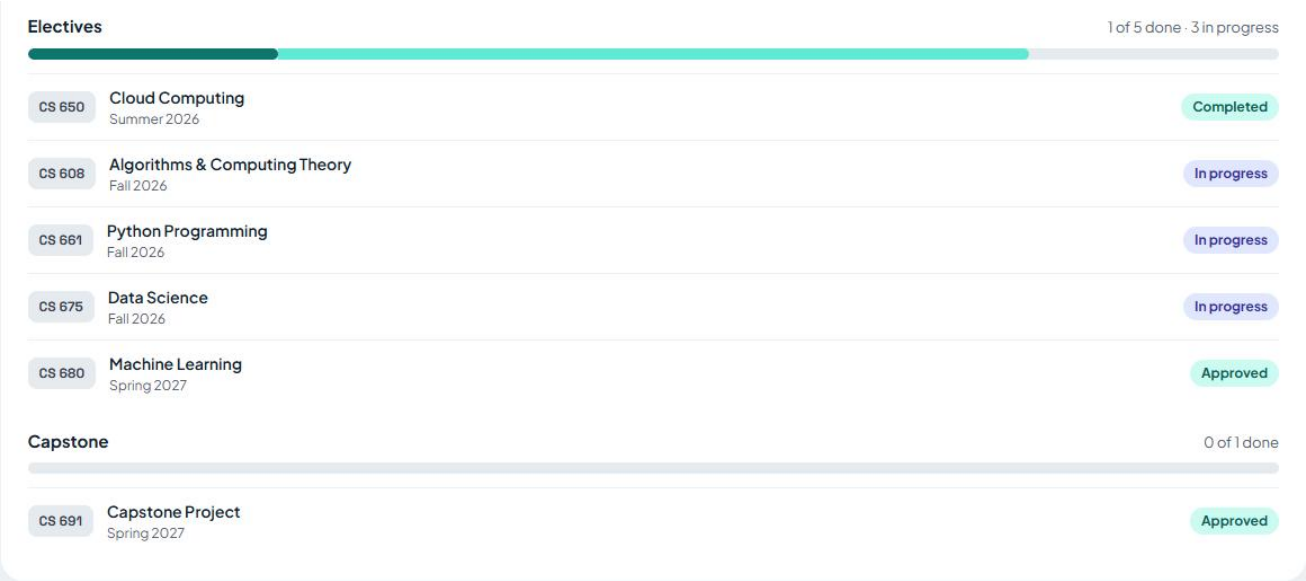


The repaired name search returns the matching advisee.

Open the student record or Approvals and review the request before choosing Approve or Request changes. The checked request is 1709. Refresh the student Degree plan after approval. Both course rows should show Approved. The persisted result is the evidence that the student and advisor views share the same saved decision.

For repeated demonstrations, use Reopen in the advisor workflow while the action is available. Refresh the student view and confirm both rows have returned to Pending approval. Do not reset the whole database to repeat one approval.

3 Confirm the result and handle unavailable data



The Spring 2027 course plan after approval and student refresh.

If a plan still looks unchanged, confirm that both browser sessions use the same API and database, then refresh the page. Check that the advisor owns the student record. A denied request or an unavailable service should be investigated through the API response rather than treated as a saved decision.

Academic risk research demonstration

The optional research interface runs separately on port 55175 with API port 58002 in the existing validation environment. It is not part of the main application installation. Finley Brook, student 10016, has two active enrollment records in the synthetic demonstration.

The original model and calibration files are absent. The prediction endpoint currently returns HTTP 503 with RESEARCH_DEMO_UNAVAILABLE, and the interface displays Not scored. Missing coverage leaves the probability null and can display Unknown data. Read the descriptive grade or engagement records on their own terms. They are not a risk probability or a SHAP explanation.

To restore inference, install the original model and calibration files, verify their SHA-256 hashes against ml/configs/FINAL_EXPERIMENT_V1.json and rerun the artifact integration tests.

4 Other application pages

Student pages

My courses provides current, completed and planned courses and grade details. Schedule combines class, meeting and personal event entries. Assignments can be filtered by course and status. Messages contains permitted conversations and system notices. Book a meeting lets a student select an available staff slot. Settings contains profile, data preferences, export and account controls.

Advisor pages

The advisor dashboard summarizes the caseload and pending work. Advisees leads to each student record. Approvals contains plan and form requests. Meetings and Messages support follow up with students. Reports and Settings provide the remaining role specific controls. These routes exist in main. The browser demonstration concentrates on search, the student record and plan approval.

Administrator pages

Use Overview for platform activity and advisor workload. Users manages invites, imports and account status. Advisors manages capacity and assignment. Catalog saves drafts before publication. Audit log records changes. AI settings controls the engine and allowed context. Integrations and institution Settings hold service and policy configuration.

Catalog publication, account deactivation, advisor reassignment and account deletion change shared data. Review the target and expected effect before confirming. The course regression used isolated synthetic records.

AI and external services

The AI advisor uses permitted database context and a rule-based fallback. Sensitive topics can lead to a human meeting flow. Claude responses require a configured Claude service. Google sign-in, SMTP delivery and calendar synchronization require configured providers and have not passed external integration acceptance.

In the default demo, password reset and invite links are printed to the server log. Five wrong passwords can lock an email for 15 minutes. A deactivated account cannot sign in. If a page returns 401, sign in again. If it returns 403, check the account role and student assignment.

5 Install a local development environment

Use Python 3.10 or newer, Node.js 22.12 or newer in the 22 release line, npm and Docker Desktop. Clone https://github.com/htmw/F2026-Team3.git or extract the source archive, then open a PowerShell terminal in the project folder.

Create a disposable development database

Choose a local password and substitute it consistently in the Docker command and environment file. The example uses port 5433. If a container with this name already exists, inspect or start that container instead of creating another one.

```
docker run -d --name acadmate-fastapi-db `
  -e POSTGRES_USER=acadmate `
  -e POSTGRES_PASSWORD=CHOOSE_LOCAL_PASSWORD `
  -e POSTGRES_DB=acadmate -p 5433:5432 `
  -v acadmate-fastapi-pg:/var/lib/postgresql/data `
  postgres:16-alpine
```

Configure and seed the API

```
Set-Location backend
Copy-Item .env.example .env
py -3 -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
```

Edit backend/.env before continuing. Keep the test URL on a different database. Set ACADMATE_TODAY to reproduce the seeded October demonstration. Leave provider keys empty until you have a service configuration to test.

```
DATABASE_URL=postgresql://acadmate:CHOOSE_LOCAL_PASSWORD@localhost:5433/acadmate
TEST_DATABASE_URL=postgresql://acadmate:CHOOSE_LOCAL_PASSWORD@localhost:5433/acadmate_test
ACADMATE_TODAY=2026-10-01
FRONTEND_ORIGIN=http://localhost:5173
COOKIE_SECURE=false
DEMO_MODE=false
```

The reset command below deletes every table in DATABASE_URL. Run it only after confirming that this URL names the disposable local development database.

```
.\.venv\Scripts\python.exe test_db.py
.\.venv\Scripts\python.exe reset_db.py --yes
.\.venv\Scripts\python.exe seed_data.py
.\.venv\Scripts\python.exe -m uvicorn main:app --reload --port 8000
```

6 Start the browser and verify the installation

Keep the API terminal open. In a second PowerShell terminal, open the extracted project folder and configure the frontend proxy.

```
Set-Location frontend
npm ci
@'
VITE_API_URL=/api/v1
API_PROXY_TARGET=http://localhost:8000
'@ | Set-Content -Encoding UTF8 .env.local
npm run dev -- --host localhost --port 5173
```

Open http://localhost:5173 and sign in as Amara. The API health route at http://localhost:8000/api/v1/health should return a successful status. Open http://localhost:8000/docs to inspect the API. Verify the student degree progress, advisor search and approval sequence described earlier.

Run backend checks on a separate database

Keep Vite running. Open a third PowerShell terminal in the extracted project folder for these checks.

```
docker exec acadmate-fastapi-db createdb -U acadmate acadmate_test
Set-Location backend
.\.venv\Scripts\python.exe -m pytest -q
```

Run createdb once. If the database already exists, keep it and proceed. Pytest reads TEST_DATABASE_URL and drops and recreates its tables on every run. Confirm that it does not point to the development or any persistent database. The 7 November reference result is 353 passed.

From frontend, run npm run build to create dist.